

**Course Code: 102**  
**Course Title: MATHEMATICS**

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| Course Code                                                             | 102                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |      |      |      |      |      |      |      |      |  |
| Course Title                                                            | Mathematics<br>(Multi-Disciplinary Course – 01)<br>[Title of the course will be the one selected by the student from courses offered by college/institute out of the course basket offered by the University under the Multi-Disciplinary courses or Inter-disciplinary courses.]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |      |      |      |      |      |      |      |      |  |
| Credits                                                                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |      |      |      |      |      |      |      |      |  |
| Course Category                                                         | Multidisciplinary Course (MC-01)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |      |      |      |      |      |      |      |      |  |
| Level of Course                                                         | 100-199 ( Foundation / Introductory )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |      |      |      |      |      |      |      |      |  |
| Teaching per Week                                                       | 4 Hrs.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |      |      |      |      |      |      |      |      |  |
| Minimum weeks per Semester                                              | 15 (Including class work, examination, preparation etc.)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |      |      |      |      |      |      |      |      |  |
| Review / Revision                                                       | 2022-2023                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |      |      |      |      |      |      |      |      |  |
| Implementation Year:                                                    | A.Y. 2023-2024                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |      |      |      |      |      |      |      |      |  |
| Purpose of Course                                                       | To impart fundamental knowledge and develop mathematical abilities relevant to applications relevant to Computer Applications.<br>[Student will opt any one course of multi-disciplinary nature from other than the computer Science and Application faculty. The course will be offered by the institute/college passed by the Board of Studies of University faculties other than the computer science and application faculty.]                                                                                                                                                                                                                                                                                                                                                                                                                                                    |      |      |      |      |      |      |      |      |  |
| Course Objective                                                        | To Provide a foundation in mathematical concepts and methods that are relevant to Computer Applications and develop the ability to apply mathematical knowledge and techniques to solve problems in computing.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |      |      |      |      |      |      |      |      |  |
| Pre-requisite                                                           | Knowledge of Fundamentals of Mathematics of 10 <sup>th</sup> Grade Level                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |      |      |      |      |      |      |      |      |  |
| Course Outcomes                                                         | <b>CO1:</b> Define and explain the fundamental concepts of Mathematical Abilities in organizations.<br><b>CO2:</b> Students can apply set theory concepts to real-world scenario, such as analyzing survey data.<br><b>CO3:</b> Enhance student’s logical reasoning to solve problems in various contexts, such as puzzles or legal arguments by learning Truth table.<br><b>CO4:</b> Course aims to equip students with the knowledge and skills to define and operate matrices, compute solutions to business problems through the use of mathematical concepts and techniques.<br><b>CO5:</b> Course aims to develop students' ability to think logically and critically, as well as to apply mathematical concepts and techniques to real-world problems.<br><b>CO6:</b> Develop independent learning skills, including the ability to research and explore mathematical concept. |      |      |      |      |      |      |      |      |  |
| Mapping between Course Outcomes(CO) with Program Specific Outcomes(PSO) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | PSO1 | PSO2 | PSO3 | PSO4 | PSO5 | PSO6 | PSO7 | PSO8 |  |
|                                                                         | CO1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |      |      |      |      |      |      |      |      |  |
|                                                                         | CO2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |      |      |      |      |      |      |      |      |  |
|                                                                         | CO3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |      |      |      |      |      |      |      |      |  |
|                                                                         | CO4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |      |      |      |      |      |      |      |      |  |
|                                                                         | CO5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |      |      |      |      |      |      |      |      |  |
|                                                                         | CO6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |      |      |      |      |      |      |      |      |  |
| Course Outcome                                                          | After studying the course, students will be able to Implement acquired skills in writing codes using programming languages.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |      |      |      |      |      |      |      |      |  |

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| <b>Course Content</b>       | <p><b>Unit 1. Set Theory</b></p> <p>1.1.Introduction<br/>1.2.Representation<br/>1.3.Operation and its properties<br/>1.4.Venn Diagram<br/>1.5.Cartesian product and graph</p> <p><b>Unit 2. Functions</b></p> <p>2.1.Definition<br/>2.2.Types – Domain and Range<br/>2.3.Construction and functions</p> <p><b>Unit 3. Mathematical Logic</b></p> <p>3.1.Introduction to logic<br/>3.2.Truth Table</p> <p><b>Unit 4. Boolean Algebra</b></p> <p>4.1Definition &amp; Examples of Boolean Algebra<br/>4.2Boolean Functions<br/>4.3Representation and minimization of Boolean Functions<br/>4.4Design example using Boolean algebra</p> <p><b>Unit 5. Matrices and Determinants</b></p> <p>5.1.Matrices of order <math>M \times N</math><br/>5.2.Row and Column transformation<br/>5.3.Addition, Subtraction and multiplication of Matrices<br/>5.4.Computation of Inverse<br/>5.5.Cramer's Rule<br/>5.6.Business Application of Matrices</p> |
| <b>Reference Books</b>      | <p>1. Co-ordinate Geometry – Shanti Narayan<br/>2. Linear Algebra – Sushoma Verma<br/>3. Advanced Mathematics – B.S. Shah &amp; Co.<br/>4. Schaum's Outline of Boolean algebra and switching circuits – Elliot Mendelson<br/>5. Digital Computer Fundamentals - Tata McGraw Hill, 6th Edition, Thomas C. Bartee<br/>6. Business Mathematics - Qazi Zameeruddin, V. K. Khanna and S. K. Bhambri, Vikas Publishing House Pvt. Ltd.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Teaching Methodology</b> | Class Work, Discussion, Self-Study, Seminars and/or Assignments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Evaluation Method</b>    | <p>50% Internal assessment.<br/>50% External assessment.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |